

Climate Change: Key Facts and Trends

1. Global Temperature Rise

Earth's average surface temperature has risen by approximately 1.1 degrees Celsius since the late 19th century, a change driven primarily by increased concentrations of greenhouse gases such as carbon dioxide and methane in the atmosphere. Most of this warming has occurred in the past 40 years, with the seven most recent years being the warmest on record.

2. Greenhouse Gas Emissions

The burning of fossil fuels for electricity, heat, and transportation is the largest single source of global greenhouse gas emissions. Deforestation also contributes significantly, since trees that would otherwise absorb carbon dioxide are removed and often burned, releasing stored carbon back into the atmosphere.

3. Rising Sea Levels

Global sea level has risen about 21 to 24 centimeters since 1880, with roughly a third of that rise occurring in just the last two and a half decades. Sea level rise is caused mainly by thermal expansion of warming ocean water and by melting ice sheets and glaciers.

4. Extreme Weather Events

Scientific studies link rising global temperatures to an increase in the frequency and severity of extreme weather events, including heat waves, heavy rainfall, and stronger tropical storms. These events pose growing risks to agriculture, infrastructure, and public health worldwide.

5. Mitigation Strategies

Common strategies to mitigate climate change include transitioning to renewable energy sources such as solar and wind, improving energy efficiency, protecting and restoring forests, and developing carbon capture technologies. International agreements such as the Paris Agreement aim to coordinate global action to limit warming.